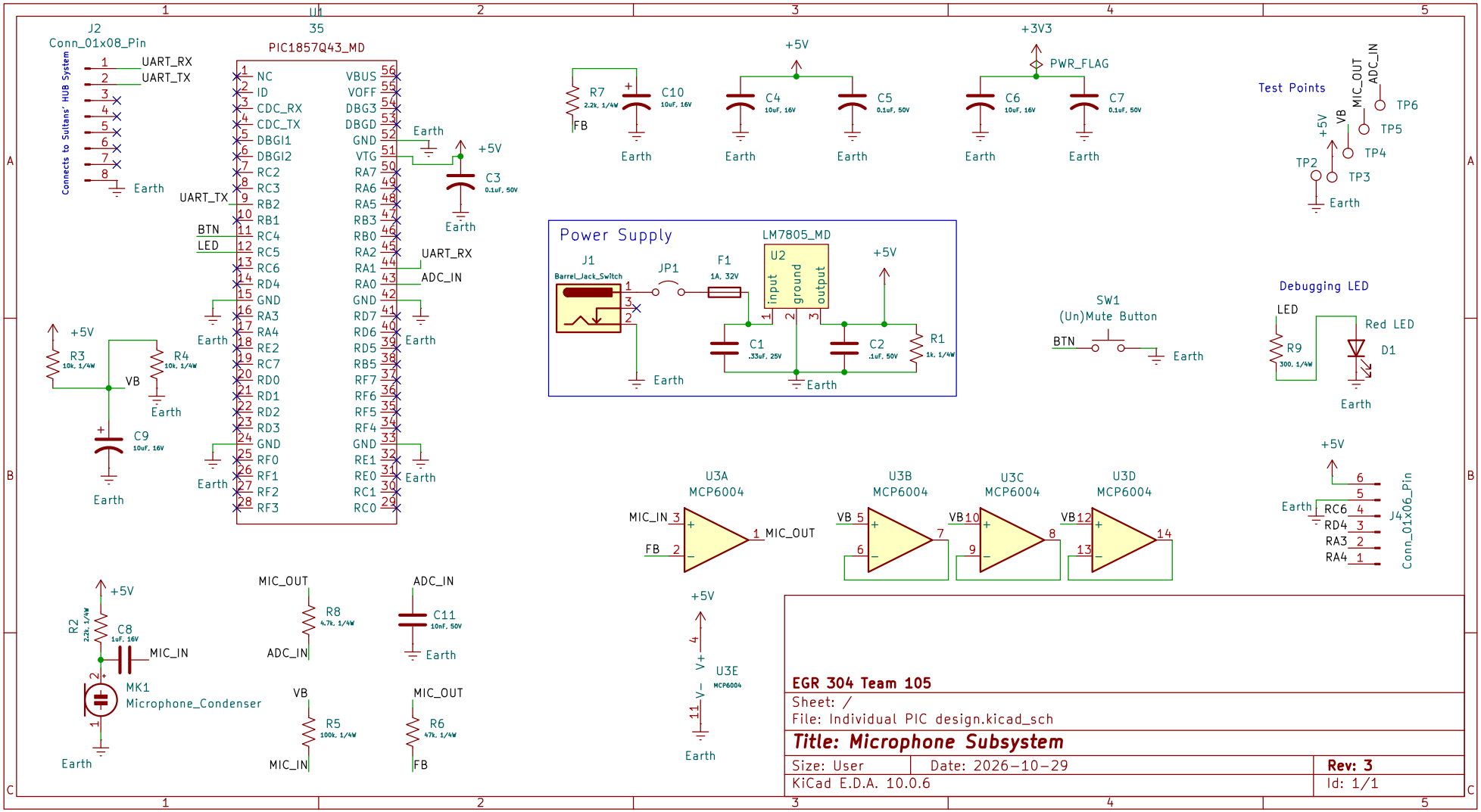
J2
Conn_01x08_Pin
Connects to Sultans' HUB System
UART_RX
UART_TX
Earth
U1
35
PIC1857Q43_MD
NC
ID
CDC_RX
CDC_TX
DBGI1
DBGI2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0
RF1
RF2
RF3
VBUS
VOFF
DBG3
DBGD
GND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0
UART_TX
BTN
LED
UART_RX
ADC_IN
+5V
C3
0.1uF, 50V
R7
2.2k, 1/4W
FB
C10
10uF, 16V
C4
10uF, 16V
C5
0.1uF, 50V
+3V3
PWR_FLAG
C6
10uF, 16V
C7
0.1uF, 50V
Test Points
MIC_OUT
ADC_IN
TP6
TP5
TP4
TP3
TP2
VB
Power Supply
J1
Barrel_Jack_Switch
JP1
F1
1A, 32V
LM7805_MD
U2
input
ground
output
C1
.33uF, 25V
C2
.1uF, 50V
R1
1k, 1/4W
SW1
(Un)Mute Button
BTN
Debugging LED
LED
R9
300, 1/4W
Red LED
D1
R3
10k, 1/4W
R4
10k, 1/4W
VB
C9
10uF, 16V
U3A
MCP6004
MIC_IN
FB
MIC_OUT
U3B
MCP6004
U3C
MCP6004
U3D
MCP6004
J4
Conn_01x06_Pin
RC6
RD4
RA3
RA4
R2
2.2k, 1/4W
C8
1uF, 16V
MK1
Microphone_Condenser
R8
4.7k, 1/4W
C11
10nF, 50V
R5
100k, 1/4W
R6
47k, 1/4W
U3E
MCP6004
V+
V-
EGR 304 Team 105
Sheet: /
File: Individual PIC design.kicad_sch
Title: Microphone Subsystem
Size: User
Date: 2026-10-29
Rev: 3
KiCad E.D.A. 10.0.6
Id: 1/1